Doping Stability Report -- Nb in LiCoO2 at 320 C

Generated: 2026-05-31 16:00   Host: LiCoO2   Dopant: Nb

Executive Summary

Nb preferentially occupies the Co layer. The formation energy difference between the two layers is +11.352 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is 0.00496 A^2/ps (stable), compared to 0.20203 A^2/ps on the Li site. The Li-site E_f is approximate due to a charge surplus of +4 that cannot be modelled in CHGNet.

Site	E_f (eV)	Charge	Compensation	MSD pass	Coord pass	Verdict
Co	+2.157	+2	Yes	PASS	FAIL	STRUCTURAL COLLAPSE
Li	+13.509	+4	Yes	FAIL	PASS	MIGRATION / UNFAVORABLE

Nb at the Co layer -- STRUCTURAL COLLAPSE

Charge situation

Nb replaces Co with a charge surplus of +2. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

Formation Energy (thermodynamic stability)

Formation energy E_f	**+2.1573 eV**	FAIL	*< 1.0 eV to pass*

E_f = +2.157 eV (approximate -- charge surplus not modelled in CHGNet). This value is large and positive, indicating that Nb incorporation is thermodynamically unfavorable under equilibrium conditions at the Co site.

Molecular Dynamics Stability (320 C, 25 ps)

MSD slope (late-time)	**0.00496 A^2/ps**	PASS	*< 0.005 to pass (plateau = stable)*
Final MSD	**0.264 A^2**	----	
Max displacement	**1.093 A**	----	
Coordination (min/mean/max)	**5 / 6.0 / 8**	FAIL	*relaxed = 6, tolerance +/-1*
Mean bond length (MD vs relaxed)	**2.037 A (relaxed: 1.9**	----	
Lattice volume change	**+0.00 %**	PASS	*< +/-5% to pass*

Late-time MSD slope = 0.00496 A^2/ps. The slope is small and below the migration threshold -- the dopant is site-stable. Final MSD = 0.264 A^2; maximum displacement from starting position = 1.09 A.

Coordination fluctuated between 5 and 8 (mean 6.0), deviating by more than +/-1 from the relaxed value (6). This may indicate structural distortion, or that the bond-length cutoff is too close to the actual bond length -- check the coordination_cutoff_A setting. Mean nearest-neighbour distance during MD = 2.037 A.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STRUCTURAL COLLAPSE

The coordination criterion failed while MSD and volume both pass, suggesting a possible cutoff artifact rather than a true structural collapse. The Nb atom barely moves (MSD = 0.264 A^2) but the coordination count fluctuates because the bond length at the Co site is close to the analysis cutoff. Consider increasing coordination_cutoff_A and re-running analysis.

Nb at the Li layer -- MIGRATION / UNFAVORABLE

Charge situation

Nb replaces Li with a charge surplus of +4. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

Formation Energy (thermodynamic stability)

Formation energy E_f	**+13.5089 eV**	FAIL	*< 1.0 eV to pass*

E_f = +13.509 eV (approximate -- charge surplus not modelled in CHGNet). This value is large and positive, indicating that Nb incorporation is thermodynamically unfavorable under equilibrium conditions at the Li site.

Molecular Dynamics Stability (320 C, 25 ps)

MSD slope (late-time)	**0.20203 A^2/ps**	FAIL	*< 0.005 to pass (plateau = stable)*
Final MSD	**2.121 A^2**	----	
Max displacement	**1.923 A**	----	
Coordination (min/mean/max)	**5 / 6.0 / 7**	PASS	*relaxed = 6, tolerance +/-1*
Mean bond length (MD vs relaxed)	**2.056 A (relaxed: 2.0**	----	
Lattice volume change	**+0.00 %**	PASS	*< +/-5% to pass*

Late-time MSD slope = 0.20203 A^2/ps. The large slope indicates significant diffusion -- the dopant is migrating. Final MSD = 2.121 A^2; maximum displacement from starting position = 1.92 A.

Coordination fluctuated between 5 and 7 (mean 6.0), within the +/-1 tolerance of the relaxed value (6). This is consistent with normal thermal vibration. Mean nearest-neighbour distance during MD = 2.056 A.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: MIGRATION / UNFAVORABLE

The formation energy is favorable but Nb migrates away from the Li site during MD. The relaxed structure is not the true resting site. Inspect the MSD and trajectory to identify where the dopant ends up.

Site Preference Comparison

The Co site is preferred by 11.352 eV over the Li site. Formation energy: +2.157 eV (Co) vs +13.509 eV (Li).

Nb preferentially occupies the Co layer. The formation energy difference between the two layers is +11.352 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is 0.00496 A^2/ps (stable), compared to 0.20203 A^2/ps on the Li site. The Li-site E_f is approximate due to a charge surplus of +4 that cannot be modelled in CHGNet.

Caveats and Limitations

? **CHGNet is charge-neutral.**

The potential does not model oxidation states, polaron formation, or electron/hole localisation. Formation energies for aliovalent dopants (charge mismatch != 0) are approximate even when structural compensation defects are included.

? **No Freysoldt/Kumagai correction.**

Electrostatic finite-size corrections for charged defects require the host dielectric tensor and the DFT electrostatic potential, neither of which is available from a neural-network potential. For publication-quality E_f of charged defects, follow up with DFT using doped or pymatgen-analysis-defects.

? **Metal-rich reference state.**

Chemical potentials are computed from elemental ground-state structures. Synthesis under oxide-rich or other conditions shifts all E_f values by additive constants; the relative ordering of sites is unaffected.

? **MD timescales are short (25 ps default).**

This is sufficient to assess local site stability and rapid migration, but cannot resolve slow diffusion mechanisms. A non-plateauing MSD confirms instability; a plateau does not guarantee stability on longer timescales.

? **Energy rankings are more trustworthy than absolute values.**

CHGNet is a universal MLIP trained on r2SCAN DFT. Formation-energy errors for transition-metal oxides are typically 50-150 meV/atom. Use relative orderings (which site is preferred?) as the primary result.